Quiz – 3

Time Allowed: 30 Mins **Total Marks: 30**

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1. Multiple Choice Questions: **(10 marks)**

Choose the correct answer:

1. If we declare int arr[5] = {1,2}; then the remaining elements of the array will be:
   A) Undefined
   B) All initialized to 0
   C) Garbage values
   D) Random values
2. The statement srand(time(0)); ensures that:
   A) Random numbers will always repeat
   B) Random numbers will be different each run
   C) Random numbers will always be 0
   D) It resets the compiler
3. Which of the following correctly declares a character array to hold "Hello"?
   A) char arr[5] = "Hello";
   B) char arr[6] = "Hello";
   C) char arr[] = "Hello";
   D) Both B and C
4. What is the range of values produced by 1 + rand() % 2?
   A) 0, 1
   B) 1, 2
   C) 0–2
   D) 1–3
5. What is the output of:

   int arr[5] = {1,2,3,4,5};

   cout << arr[2];

   A) 1
   B) 2

C) 3
D) Garbage

6. Arrays in C++ are stored in:
   A) Random locations
   B) Consecutive memory locations
   C) Register memory
   D) None

7. If #define SIZE 100 is used, what happens if we later try SIZE = 50; inside main()?
   A) SIZE changes to 50
   B) Compilation error
   C) Runtime error
   D) SIZE becomes global

8. Which of the following is **true** about arrays passed to functions?
   A) They are passed by value
   B) They are passed by reference mechanism
   C) They are copied completely
   D) Cannot be passed

9. In a 2D array declared as int arr[3][4]; how many total elements are there?
   A) 7
   B) 12
   C) 10
   D) 8

10. Which of the following correctly defines a function in C++ that takes an integer array and its size as parameters?

A)
```
void process(int arr, int size) {
   for (int i = 0; i < size; i++) {
      cout << arr[i];
   }
}
```

B)
```
void process(int arr[], int size) {
   for (int i = 0; i < size; i++) {
      cout << arr[i];
```

```
    }
  }
  C) void process(int* arr, int size) {
    for (int i = 0; i < size; i++) {
      cout << arr[i];
    }
  }
  D) void process(int arr[][], int size) {
    for (int i = 0; i < size; i++) {
      cout << arr[i];
    }
  }
```

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2. Write the output of the following code snippet: **(5 marks)**

```
#include <iostream>
using namespace std;
int main() {
  int a[3][3] = {{1,2,3}, {4,5,6}, {7,8,9}};
  int sum = 0;
  for (int i = 0; i < 3; i++) {
    for (int j = 2; j >= 0; j--) {
      if ((i + j) % 2 == 0)
        sum += a[i][j];
      else
        sum -= a[i][j];
    }
```

```
    }
    cout << sum;
    return 0;
}
```

OUTPUT:

3.Dry run the code snippet and write the state of array at every iteration and write output at last: (3+2 marks)

```
#include <iostream>
using namespace std;
// Function to update array elements
void modifyArray(int arr[], int size) {
    for (int i = 0; i < size; i++) {
        arr[i] = arr[i] + (i % 3);
        for (int j = 0; j < i; j++) {
            arr[j] = arr[j] - 1;
        }
    }
```

```
}
int main() {
    int arr[6] = {2, 4, 6, 8, 10, 12};
    modifyArray(arr, 6);
    cout << "Array after modification: ";
    for (int i = 0; i < 6; i++) {
        cout << arr[i] << " ";
    }
    return 0;
}
```

OUTPUT:

Q5. Calculate determinant of 2x2 matrix **(10 marks)**

Write a C++ program that calculates the **determinant** of a **2×2 matrix** using a **function**.
You must use:

- #define SIZE 2 to declare the size of the matrix.
- A function **int determinant(int matrix[SIZE][SIZE])** that calculates and returns the determinant.
- The determinant of a 2×2 matrix.

$$\det = (a_{00} \cdot a_{11}) - (a_{01} \cdot a_{10})$$